Project Overview & System Architecture

1. Project Title & Purpose

Smart Pill Box Reminder & Medication Safety Monitor

The Smart Pill Box is an embedded healthcare assistance device built on the **STM32F103C8T6 (ARM Cortex-M3)** microcontroller. Its primary purpose is to:

1. Provide accurate, real-world scheduled medication alarms for patients taking multiple daily doses.

2. Ensure patient adherence through audio-visual reminders (SSD1306 OLED, LEDs, Piezo buzzer).

3. Prevent dangerous medication errors using an active verification system ("Wrong Pill" alert mechanism).

4. Allow complete offline configuration through a simple 3-button human-machine interface (HMI).

5. Maintain long-term real time using a battery-backed DS1307 Real-Time Clock module over I2C with seamless software tick fallback.

6. Provide intake event audit logging (**`LOGS.CSV`**) and schedule configuration persistence (**`CONFIG.TXT`**) via a **MicroSD Card Module over SPI1**.

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2. Hardware Architecture & Block Diagram

```
graph TD
    subgraph Controller [STM32F103C8T6 Blue Pill]
        I2C1[I2C1 Peripheral: PB6 SCL, PB7 SDA]
        SPI1[SPI1 Peripheral: PA4 CS, PA5 SCK, PA6 MISO, PA7 MOSI]
        GPIO_IN[Push Buttons: PB12, PB14, PB15]
        GPIO_OUT[Indicators: PB1 LED1, PB10 LED2, PB0 Buzzer]
        SYS_TICK[HAL SysTick / Timer: 1ms]
    end

    subgraph I2C_Bus [Shared I2C Bus - 100 kHz]
        OLED[SSD1306 128x64 OLED Display - Addr: 0x3C / 0x78]
        RTC[DS1307 Real-Time Clock Module - Addr: 0x68 / 0xD0]
    end

    subgraph SPI_Storage [SPI Storage Bus - 9 MHz]
        SD[MicroSD Card Module: CONFIG.TXT & LOGS.CSV]
    end

    subgraph Inputs [User Input Controls]
        BTN1[Button 1: Dose 1 Confirm / Menu Increment]
        BTN2[Button 2: Dose 2 Confirm / Menu Select]
        BTN3[Button 3: Setup Menu / Exit]
    end

    subgraph Outputs [Audio-Visual Feedback]
        LED1[Pill 1 Green LED]
        LED2[Pill 2 Yellow LED]
        BUZZ[Piezo Buzzer]
    end

    subgraph Power [Power Distribution]
        P3V3[3.3V Rail: STM32, OLED]
        P5V[5V Rail: DS1307 VCC, MicroSD VCC]
        BATT[CR2032 3V Coin Cell: DS1307 Backup]
    end

    I2C1 <==> I2C_Bus
    SPI1 <==> SPI_Storage
    GPIO_IN <== Inputs
    GPIO_OUT ==> Outputs
    Power -.-> Controller
    Power -.-> I2C_Bus
    Power -.-> SPI_Storage
```

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3. System Wiring & Pinout Table

Peripheral	Signal	STM32 Pin	Mode	Purpose
I2C1	SCL	**PB6**	AF Open-Drain + External Pullup	Clock line for OLED & DS1307
I2C1	SDA	**PB7**	AF Open-Drain + External Pullup	Data line for OLED & DS1307
SPI1	CS	**PA4**	GPIO Push-Pull Output	SD Card Chip Select
SPI1	SCK	**PA5**	Alternate Function Push-Pull	SPI Clock (up to 9 MHz)
SPI1	MISO	**PA6**	GPIO Input (Internal Pull-Up)	Serial data from SD to MCU

SPI1	MOSI	**PA7**	Alternate Function Push-Pull	Serial data from MCU to SD
Button 1	Input	**PB12**	GPIO Input (Internal Pull-Up)	Pill 1 OK / Value Increment (+)
Button 2	Input	**PB14**	GPIO Input (Internal Pull-Up)	Pill 2 OK / Menu Item Select / Next
Button 3	Input	**PB15**	GPIO Input (Internal Pull-Up)	Setup Menu Open / Exit
Pill 1 LED	Output	**PB1**	GPIO Push-Pull Output	Pill 1 Visual Reminder Indicator
Pill 2 LED	Output	**PB10**	GPIO Push-Pull Output	Pill 2 Visual Reminder Indicator
Buzzer	Output	**PB0**	GPIO Push-Pull Output	Audible Frequency Alert
ST-Link v2	SWDIO	**PA13**	Serial Wire Data IO	SWD Debug / Flashing
ST-Link v2	SWCLK	**PA14**	Serial Wire Clock	SWD Debug / Flashing

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4. Software State Machine Workflow

```
stateDiagram-v2
    [*] --> SPLASH: Power-On Reset
    SPLASH --> NORMAL_IDLE: 1.2s delay & I2C Init

    state NORMAL_IDLE {
        [*] --> Monitoring
        Monitoring --> Monitoring: Poll DS1307 every 500ms
    }

    NORMAL_IDLE --> REMINDER_DOSE_1: Time == Dose 1 Alarm
    NORMAL_IDLE --> REMINDER_DOSE_2: Time == Dose 2 Alarm
    NORMAL_IDLE --> SETUP_MENU: Button 3 Pressed

    state REMINDER_DOSE_1 {
        [*] --> Alerting_1: Normal Blink 500ms
        Alerting_1 --> Wrong_Button_1: Button 2 Pressed (Wrong)
        Wrong_Button_1 --> Wrong_Button_1: Fast Blink 150ms + Warning Screen
    }

    state REMINDER_DOSE_2 {
        [*] --> Alerting_2: Normal Blink 500ms
        Alerting_2 --> Wrong_Button_2: Button 1 Pressed (Wrong)
        Wrong_Button_2 --> Wrong_Button_2: Fast Blink 150ms + Warning Screen
    }

    REMINDER_DOSE_1 --> DOSE_CONFIRMED: Button 1 Pressed (Correct)
    REMINDER_DOSE_2 --> DOSE_CONFIRMED: Button 2 Pressed (Correct)

    DOSE_CONFIRMED --> NORMAL_IDLE: 2s Success Display

    state SETUP_MENU {
        [*] --> Navigating: B1=Cycle, B2=Choose
    }

    SETUP_MENU --> NORMAL_IDLE: Button 3 Pressed
    SETUP_MENU --> SETUP_EDIT_HOUR: Button 2 Pressed
    SETUP_EDIT_HOUR --> SETUP_EDIT_MINUTE: Button 2 Pressed (B1 increments)
    SETUP_EDIT_MINUTE --> SETUP_CONFIRM: Button 2 Pressed (B1 increments)
    SETUP_CONFIRM --> SETUP_FEEDBACK: Button 2 Confirms (B1 toggles Save/Cancel)
    SETUP_FEEDBACK --> NORMAL_IDLE: 1.5s Display Delay
```

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5. Potential Evaluator Questions & Key Answers

- **Q: Why are both the OLED and RTC connected to the same pins (`PB6` and `PB7`)?**
- *A: I2C is a shared bus protocol. Each peripheral has a distinct 7-bit hardware address (OLED: `0x3C`, RTC: `0x68`). The master initiates transfers by broadcasting the address, allowing multiple slaves on two wires.*
- **Q: What happens if the power cuts off?**
- *A: The DS1307 module has a CR2032 coin battery backup, ensuring real time ticks continuously. When power restores, the STM32 immediately polls the RTC and retrieves the accurate wall-clock time.*
- **Q: How does the system prevent accidental double-dosing or multiple alarms in the same minute?**
- *A: In the alarm evaluation logic, each dose struct holds `lastTriggeredDay`. Once triggered on day `X`, the alarm won't rearm until day `X+1` or until modified in the menu.*